

CS 255 Business Requirements Document – Jaiden Forrest Logan

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- The purpose of this project is to design a system for DriverPass that helps students prepare for their DMV driving tests.
- DriverPass wants the system to give customers access to online classes, practice tests, and on-the-road driving lessons.
- The system should allow customers to create accounts, schedule driving lessons, track their progress, and access training materials.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- DriverPass sees a problem with students failing their DMV driving tests because they are not prepared enough.
- DriverPass wants to solve this problem by offering better driver training through online practice exams, online classes, and in-person driving lessons.
- The system needs to support multiple types of users, including customers, the owner, the IT officer, the secretary, and driving instructors.
- Customers need to be able to register, choose a package, schedule or modify lessons, and view online test progress.
- The secretary needs to be able to enter customer information and schedule appointments for customers who call or visit the office.
- The IT officer needs access to manage user accounts, reset passwords, block access, and maintain the system.
- The owner needs to view reports, track activity, and see who created, changed, or canceled reservations.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

- The system should allow customers to create an account and manage their own driving lesson reservations online.
- The system should allow the secretary to create, modify, and cancel appointments for customers.
- The system should track each lesson by customer, driver, car, date, time, pickup location, and drop-off location.

- The system should allow customers to choose from the three available training packages.
- The system should allow DriverPass to disable a package if it is no longer being offered.
- The system should track online practice test progress, including test name, time taken, score, and status.
- The system should record driver notes for each lesson, including lesson time, start hour, end hour, and comments.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The system should run as a web-based system that can be accessed from a computer or mobile device.
- The system should allow users to view information online, but users should only be able to modify or update data while connected to the internet to avoid duplicate or outdated data.
- The system should allow reports and some information to be downloaded so the owner can work with the data outside the system, such as in Excel.
- The system should be reliable enough for customers and employees to schedule, modify, and cancel reservations without major delays.
- The system should receive updates from the DMV when rules, policies, or sample questions change.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The system should be web-based and preferably cloud-based so DriverPass does not have to manage most backup and security issues directly.
- The system should work through a standard web browser on computers and mobile devices.
- The back end will need a database to store customer accounts, reservations, training packages, driver information, vehicle information, online test progress, reports, and activity records.
- The system should support connection to DMV updates so training material can stay current.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system should distinguish users by account type, such as customer, owner, IT officer, secretary, and driver.
- Each user role should have access only to the tools and information needed for that role.
- The system should accurately track who creates, modifies, or cancels a reservation.
- The system should match each driving lesson with the correct customer, driver, vehicle, date, time, pickup location, and drop-off location.
- The system should alert an administrator or IT officer if there is an account access issue, failed password reset issue, or unusual activity that may need attention.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should allow the IT officer to add, remove, block, and modify user accounts without changing the system code.
- The system should allow DriverPass to disable a package if it is no longer being offered.
- The system should be flexible enough to support future updates, such as adding or removing training packages in a later release.
- The system should be able to accept DMV updates so practice tests and training materials stay current.
- The IT officer should have full access to user account management and system maintenance tools.

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- The system should require users to log in with a username and password.
- The system should allow customers to reset their password automatically if they forget it.
- The IT officer should be able to reset accounts, block access, and remove access when employees leave the company.
- The system should protect customer personal information and payment information.
- The system should use secure communication between the client and server, especially when handling login credentials and payment details.
- The system should track user activity so DriverPass can see who created, changed, or canceled reservations.

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to create an account.
- The system shall allow customers to log in and access their account.
- The system shall allow customers to schedule, modify, and cancel driving lesson reservations online.
- The system shall allow the secretary to create, modify, and cancel appointments for customers who call or visit the office.
- The system shall allow customers to choose from available training packages.
- The system shall allow DriverPass to disable a training package.
- The system shall track which customer is assigned to which driver, vehicle, date, and time.
- The system shall store pickup and drop-off locations for each driving lesson.
- The system shall track online test progress, including test name, time taken, score, and status.
- The system shall store driver notes, including lesson time, start hour, end hour, and driver comments.
- The system shall allow the IT officer to manage accounts, reset passwords, and block access.
- The system shall notify DriverPass when the DMV updates rules, policies, or sample questions.
- The system shall allow the owner to print or download activity reports.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The interface should be web-based and usable on computers and mobile devices.
- Customers should be able to register, log in, view training materials, take practice tests, view progress, and manage appointments.
- The secretary should be able to enter customer information and schedule appointments for customers.
- The owner should be able to view reports and activity history.
- The IT officer should be able to manage accounts and access system maintenance tools.
- The interface should include forms for student information, contact information, lesson scheduling, online test progress, and driver notes.
- The system should include a contact page and a way for DriverPass to contact students.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Users will have access to the internet when they need to update or modify system information.
- Customers will have basic access to a computer or mobile device with a web browser.
- DriverPass will provide the training content, online class materials, and practice test questions.

- DriverPass will provide driver, vehicle, package, and scheduling information needed for the system.
- The DMV connection or update process will be available in some form so DriverPass can receive updated rules, policies, or sample questions.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The first version of the system will not allow nondevelopers to fully add or remove system modules.
- Package customization may be limited in the first release, although DriverPass should still be able to disable a package.
- Offline access will be limited to downloaded reports or information, since users should not update data while offline.
- The system design depends on the schedule and requirements given in the interview, so major changes from DriverPass could affect the timeline.
- The DMV update feature may depend on what type of access or update method the DMV provides.

Gantt Chart

DriverPass Project Schedule

Task	January	February	March	April	May	Exact Dates
Collect Requirements						Jan 22-Feb 4
Create Use Case Diagrams						Jan 11-Feb 18
Build Activity Diagrams						Feb 18-Mar 9
Research UI Designs	*			*	*	Feb 27-Mar 7
Build Class Diagram						Mar 1-Mar 9
Get Customer Approval						Mar 10-Mar 11
Build Interface	*	*		*	*	Mar 12-Mar 24
Link DB to Interface						Mar 24-Apr 3
Build Business Logic						Apr 5-Apr 27
Test System						Apr 27-May 7
Deliver System	*	*	*	*		May 8-May 9
Sign-off Meeting						May 9-May 10